

Design, Analysis and Simulation of a CMOS Voltage Controlled Oscillator

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Abstract—This paper presents the design, analysis, and simulation of a CMOS Voltage Controlled Oscillator (VCO) operating at a center frequency of 5.25 GHz. The VCO is designed using 45nm PDK models with a supply voltage of 1.2V. The oscillator achieves a tuning range greater than 10% with VCO gain greater than 900 MHz/V as well as Phase Noise at 600 kHz less than -111 dBc/Hz and Phase Noise at 1 MHz less than -118 dBc/Hz.

The design methodology, hand calculations, simulation results, and optimization techniques for power consumption are discussed in detail.

Index Terms—CMOS, 45nm technology, voltage controlled oscillator (VCO), Tuning Range, Phase Noise, Periodic Steady-State.

I. INTRODUCTION

THIS project focuses on the design of a fully-integrated CMOS VCO using 45nm technology with a center frequency of 5.25 GHz. The design must meet stringent specifications including tuning range greater than 10% with VCO gain greater than 900 MHz/V as well as Phase Noise at 600 kHz less than -111 dBc/Hz and Phase Noise at 1 MHz less than -118 dBc/Hz, all while minimizing power consumption with a 1.2V supply.

This paper is organized as follows: Section II presents the achieved specifications and simulation results. Section III describes the design methodology and hand calculations. Section IV covers power consumption optimization. Section V concludes the paper.

II. ACHIEVED SPECIFICATIONS AND SIMULATION RESULTS

A. Achieved Specifications

TABLE 1
Required Specifications and Achieved Results

Specification	Requirement	Achieved Result
V _{dd}	1.2 V	1.2 V
Center Frequency, f _c	5.25 GHz	5.25 GHz
Tuning Range	> 10 % (4.725 GHz to 5.775 GHz)	4.65 GHz to 5.79 GHz
K _{vco}	> 900 MHz/V	948 MHz/V
Phase Noise @ 600 kHz	< -111 dBc/Hz	< -114.6 dBc/Hz
Phase Noise @ 1 MHz	< -118 dBc/Hz	< -119.3 dBc/Hz
Load capacitance (from next stage)	> 80 fF	80 fF + 200fF
Power Consumption	< 20 mW	15.3 mW

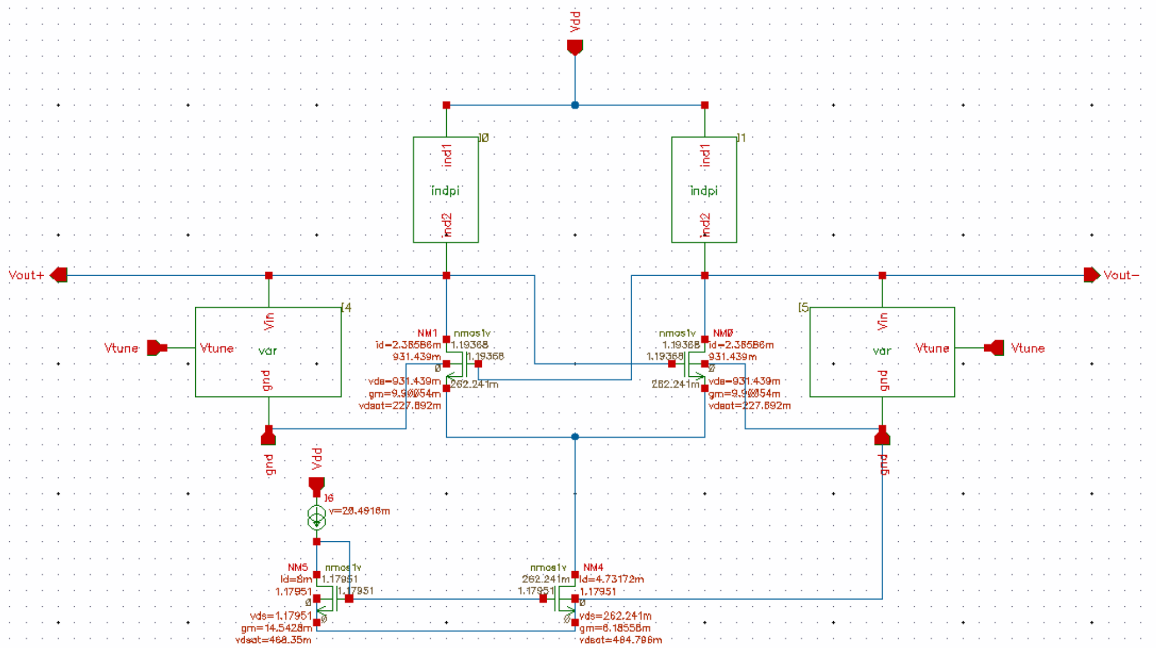


Fig 1. Voltage Controlled Oscillator Schematic

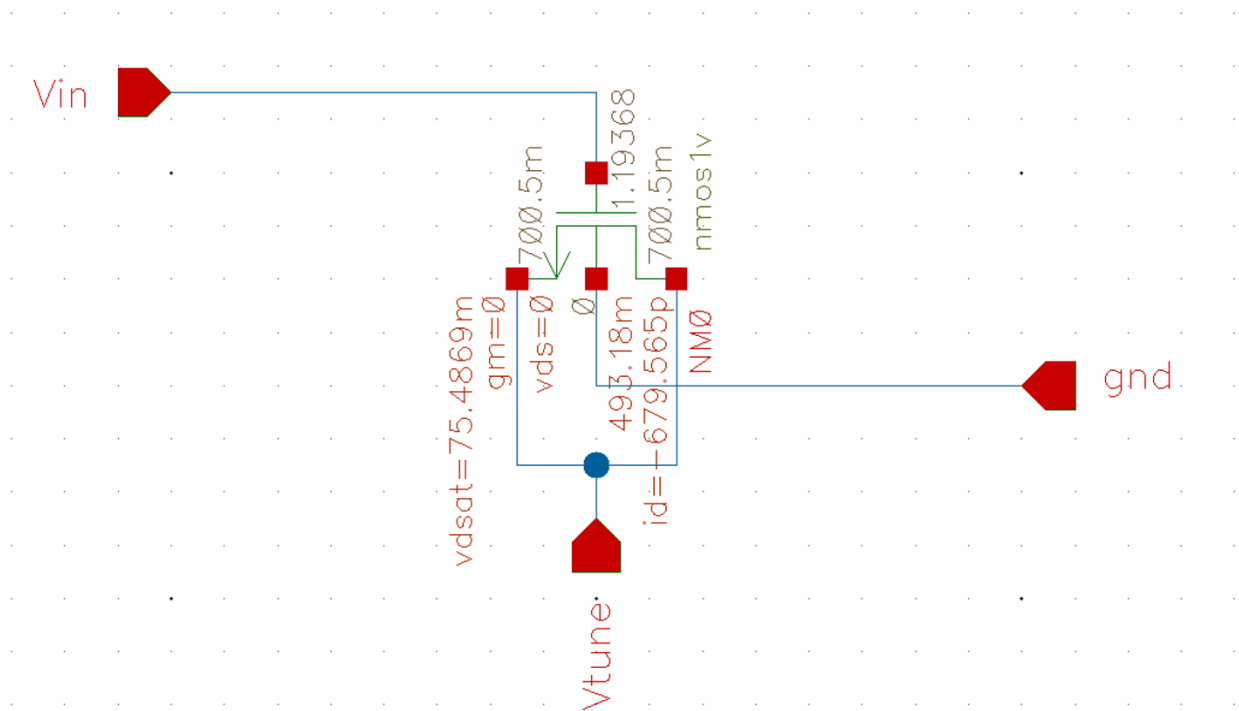


Fig 2. Varactor Schematic

The VCO was simulated using Cadence Spectre RF with 45nm PDK models with the following testbench. Transient analyses were performed for 100 ns to estimate settling time and identify power consumption. Periodic steady-state (PSS) analysis was conducted to identify the VCO tuning range and VCO gain specification. Pnoise analysis was used to identify the phase noise of the VCO as well as to identify major contributors to phase noise with a noise summary.

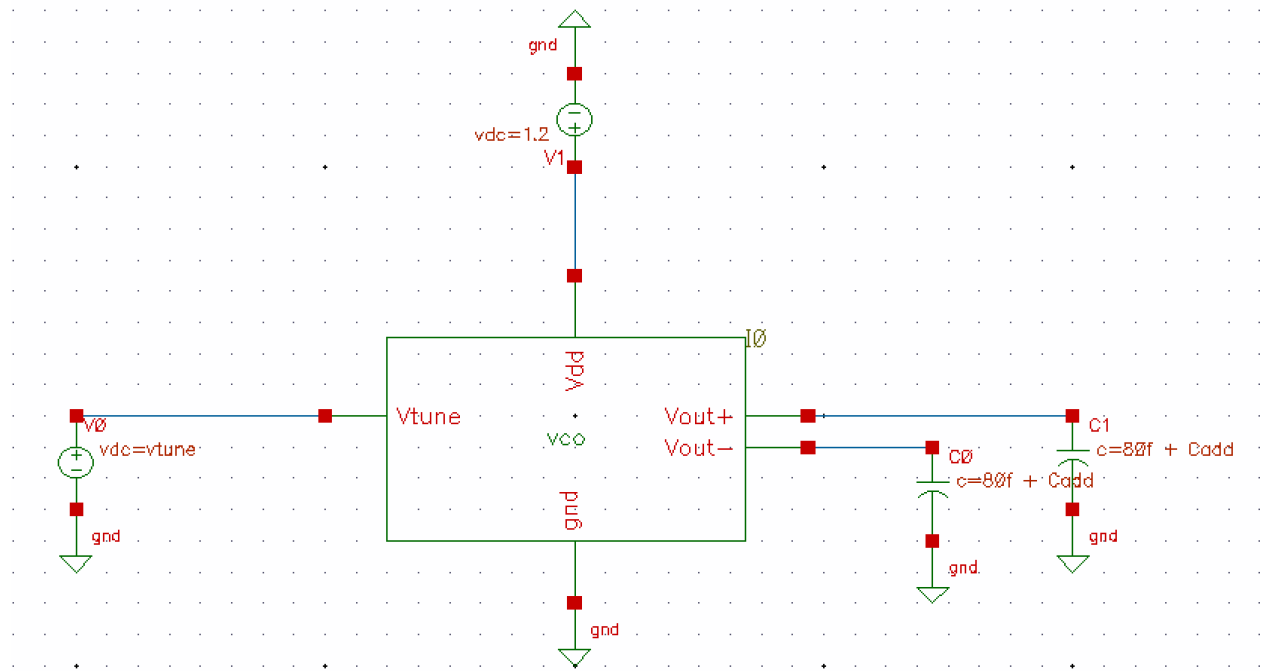
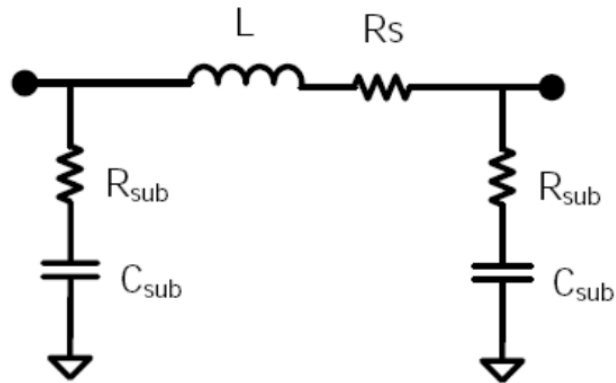


Fig 3. VCO Testbench

B. DC Operating Points

A Pi model inductor was used for the design of this VCO.



$$Rs = -0.02 \left(\frac{L}{1nH} \right)^2 + \left(\frac{L}{1nH} \right) + 1.5$$

$$Rsub = 0.0789 \left(\frac{L}{1nH} \right) + 3.4893 \left(\frac{L}{1nH} \right)^{-1} + 31.707$$

$$Csub \text{ (in pF)} = -0.0005 \left(\frac{L}{1nH} \right)^2 + 0.0312 \left(\frac{L}{1nH} \right) + 0.0543$$

Fig 4. Pi Model Inductor

The following DC operating point values were used for the components of the VCO.

TABLE 2

DC Operating Points of VCO Components

V _{dd}	1.2 V
V _{tune} @ 5.25 GHz center frequency	700.5 mV
C _{add} 0,1	200 fF
C _{load} 0,1	80 fF + C _{add} = 280 fF
L _{indpi} 0,1	1.2 nH
NM _{0,1} gm	9.90054 mS
NM _{0,1} W/L	10*(1u/45n)
NM _{0,1} V _{ds}	931.439 mV
NM _{0,1} V _{dsat}	227.692 mV
VAR _{0,1} gm	0 mS
VAR _{0,1} W/L	3.5*(10u/1u)
VAR _{0,1} V _{ds}	0 mV
VAR _{0,1} V _{dsat}	75.4869 mV (OK since should be in region 3)
NM ₄ gm	6.18558 mS
NM ₄ W/L	10*(2u/90n)
NM ₄ V _{ds}	262.241 mV
NM ₄ V _{dsat}	484.796 mV
NM ₅ gm	14.5428 mS
NM ₅ W/L	10*(2u/90n)
NM ₅ V _{ds}	1.17951 V
NM ₅ V _{dsat}	468.35 mV

C. Transient Simulation

A 100 ns transient simulation of the VCO was conducted to identify approximate settling time (< 2 ns) and calculated average power draw. The average current through the V_{dd} (1.2 V) and V_{tune} (700.5 mV) sources were found to be 12.67 mA and -11.68 uA respectively, thus the power was calculated to be 15.3 mW. Plots showing the differential and singled-ended output waveforms follow.

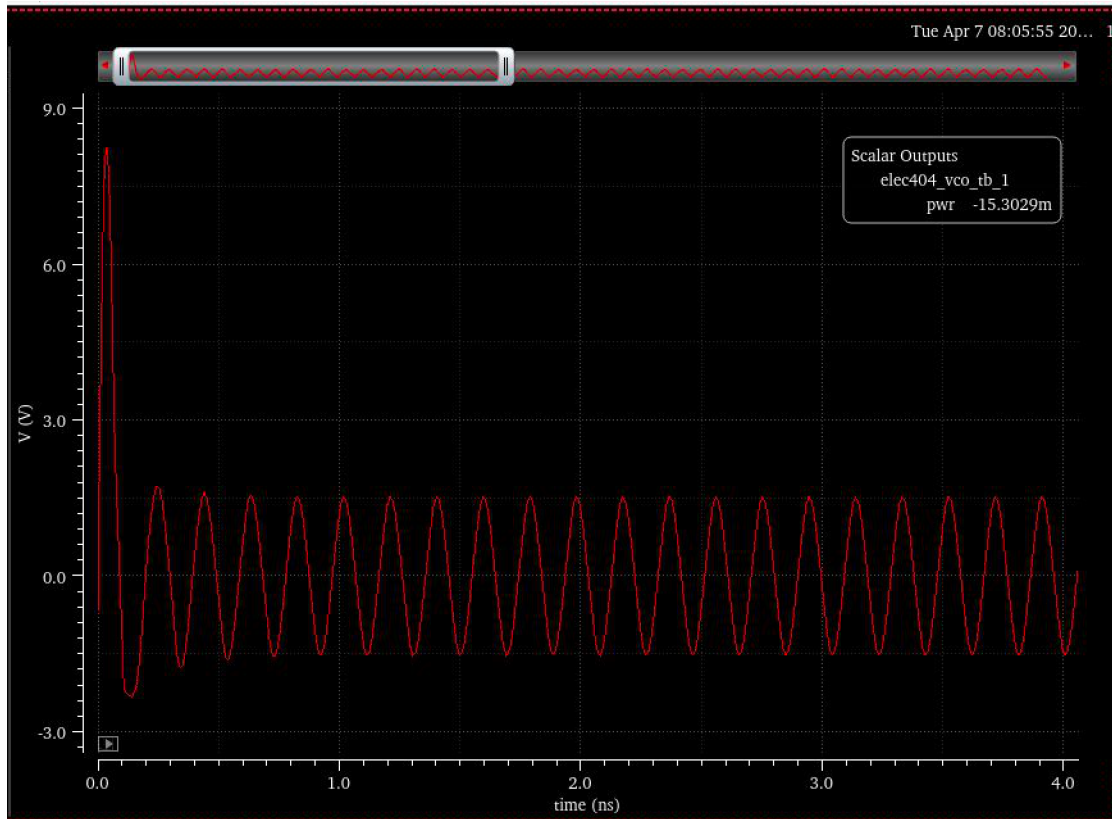


Fig 5. Differential Output Waveform (Settling Time < 2 ns)

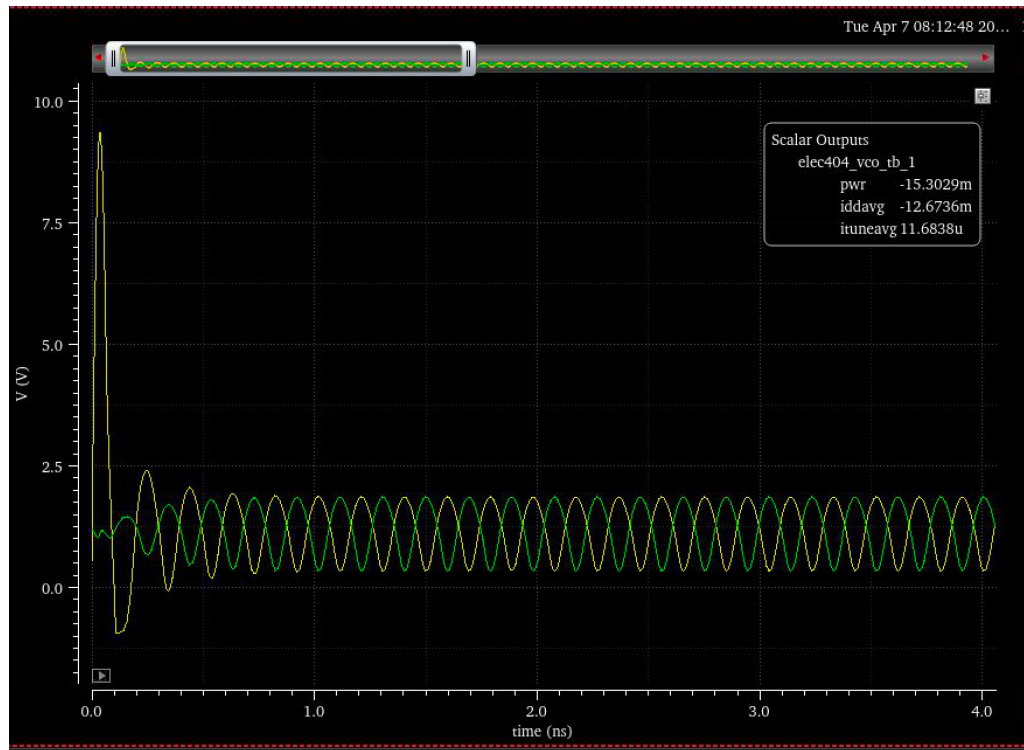


Fig 6. Single-Ended Output Waveforms (V+ yellow, V- green)

D. PSS Simulation

Periodic Steady State analyses were run in order to find the VCO's tuning range and VCO gain by sweeping V_{tune} of the varactor from 0 V to 1.2 V. A plot indicating the supply pushing response was also found by sweeping the input V_{dd} of the overall VCO from 1.1 V to 1.3 V

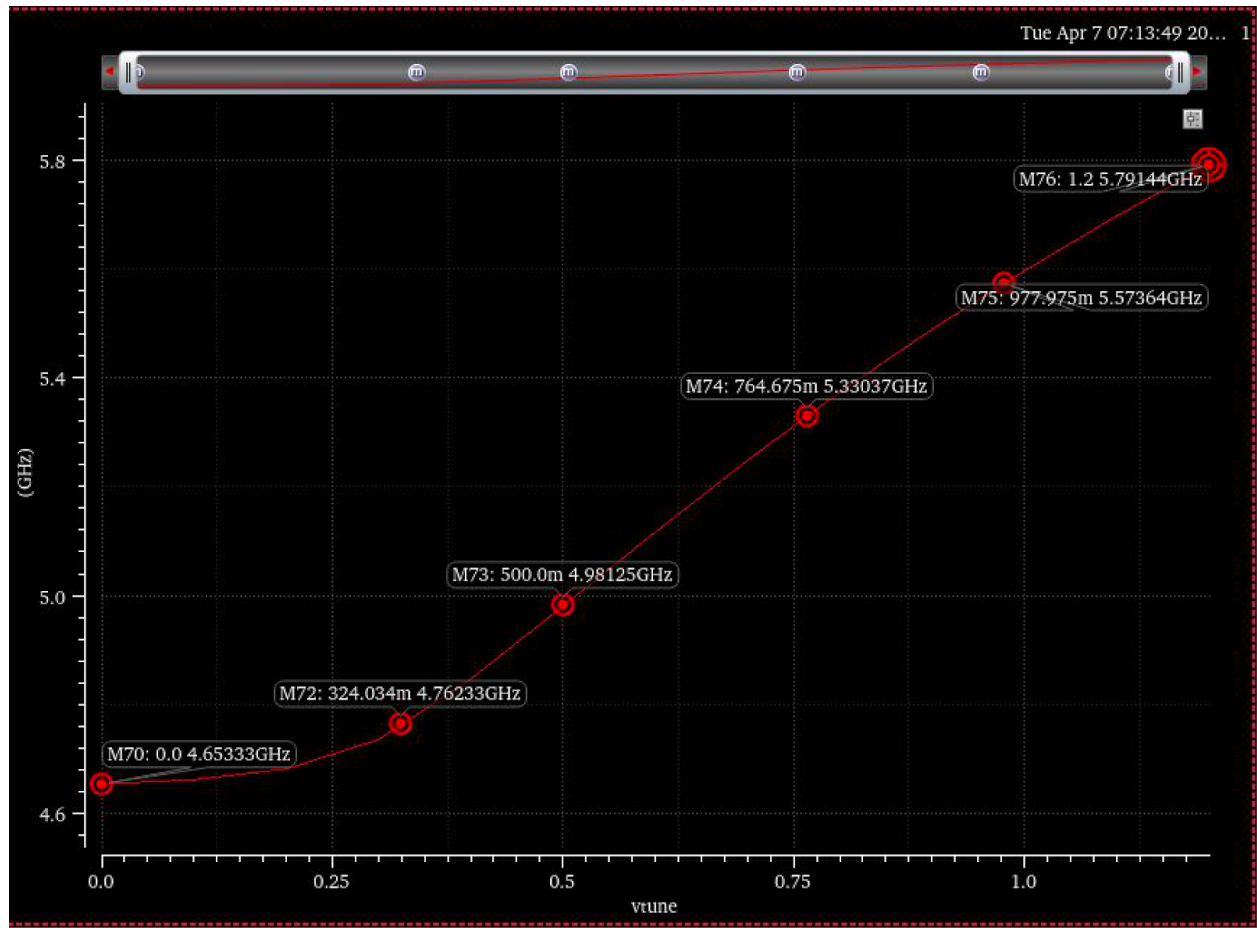


Fig 7. Differential Output Frequency vs V_{tune}

Figure 7 illustrates the differential output frequency vs V_{tune} . The total tuning range is shown to be from 0.465333 GHz to 5.79144 GHz. This corresponds to a total average K_{vco} of $(5.79144 - 4.65333) \text{ GHz} / 1.2 \text{ V} = 948.425 \text{ MHz/V}$ on average. The more linear region in the middle of the tuning range has a VCO gain of approximately $(5.57364 - 4.76233) / (0.977975 - 0.324034) = 1240.647092 \text{ MHz/V}$.

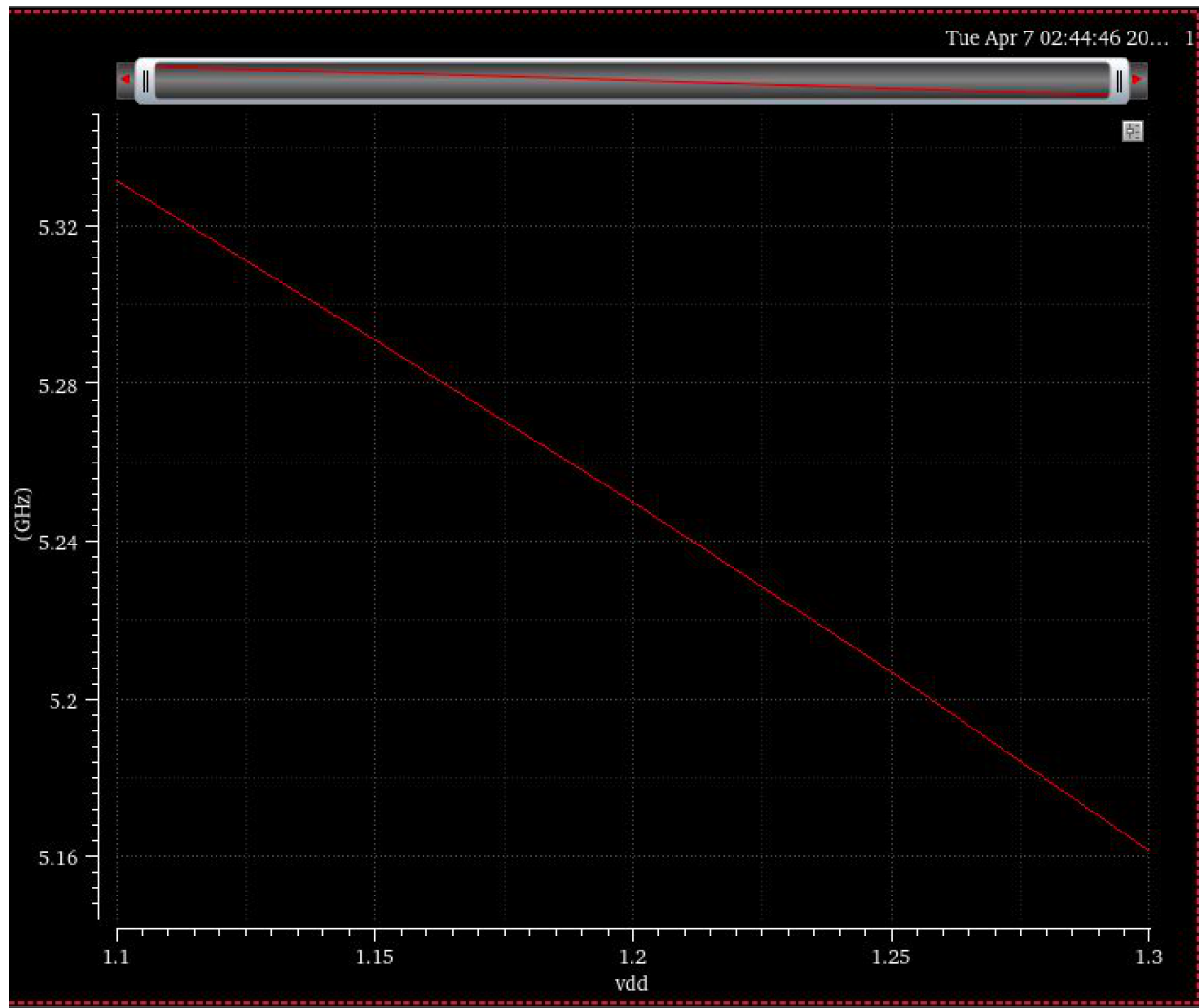


Fig 8. Supply Pushing

The plot in Figure 8 was generated by sweeping V_{dd} from 1.1 V to 1.3 V to observe the supply pushing response.

E. Pnoise Simulation

PM Pnoise simulations were conducted in order to measure the phase noise of the VCO. In Figure 9, we can see that the spot phase noise at 600 kHz and 1 MHz offsets from the 5.25 GHz center frequency meet the required specification at -114 dBc/Hz and -119 dBc/Hz respectively.

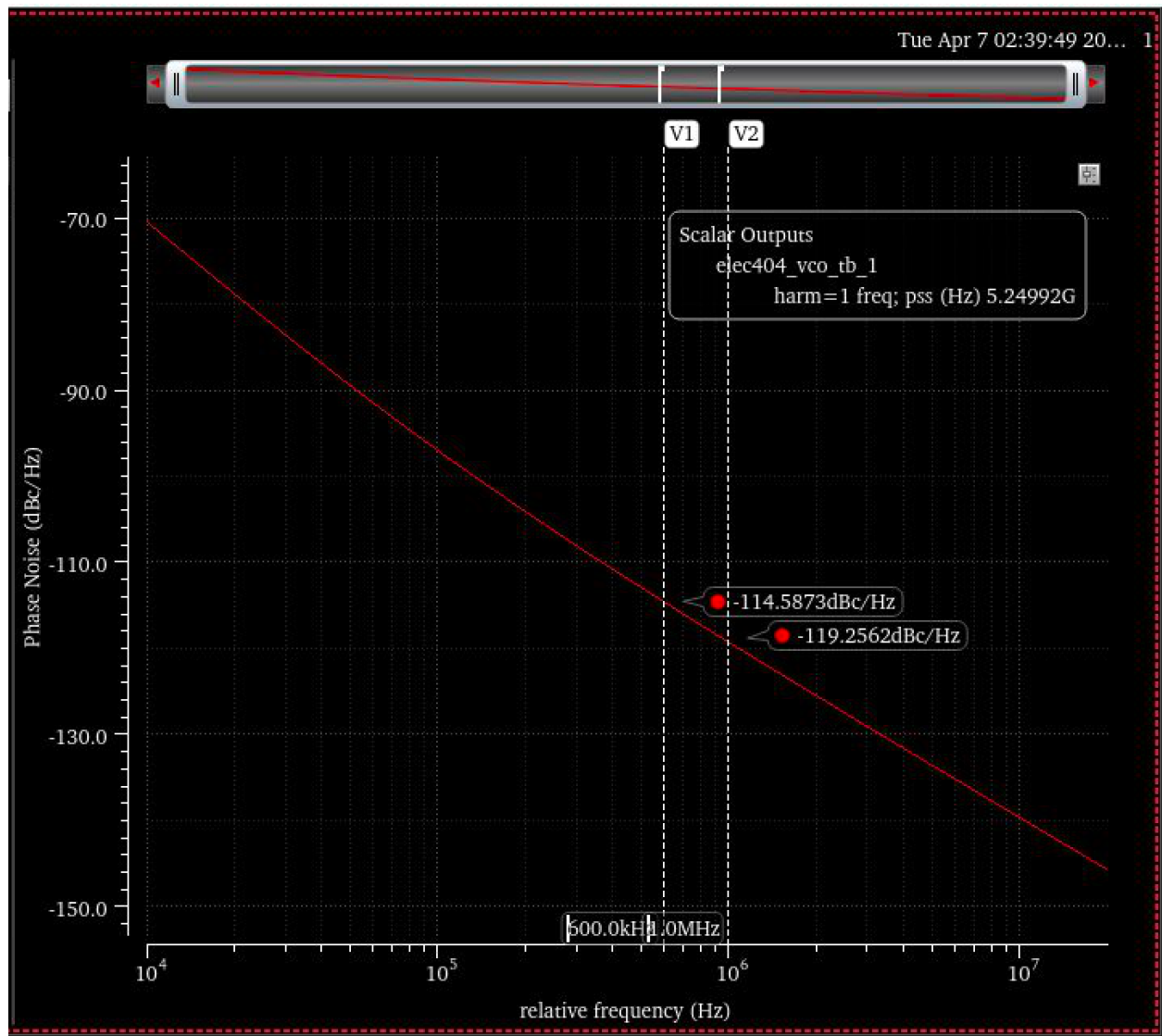


Fig 9. Phase Noise vs Relative Frequency Offset from 10 kHz to 20 MHz

We can also plot the phase noise at 600 kHz and 1 MHz offsets as a function of tuned frequency which is shown in Figure 10. The spot phase noise for the offsets relative to 5.25 GHz is indicated with a vertical marker.

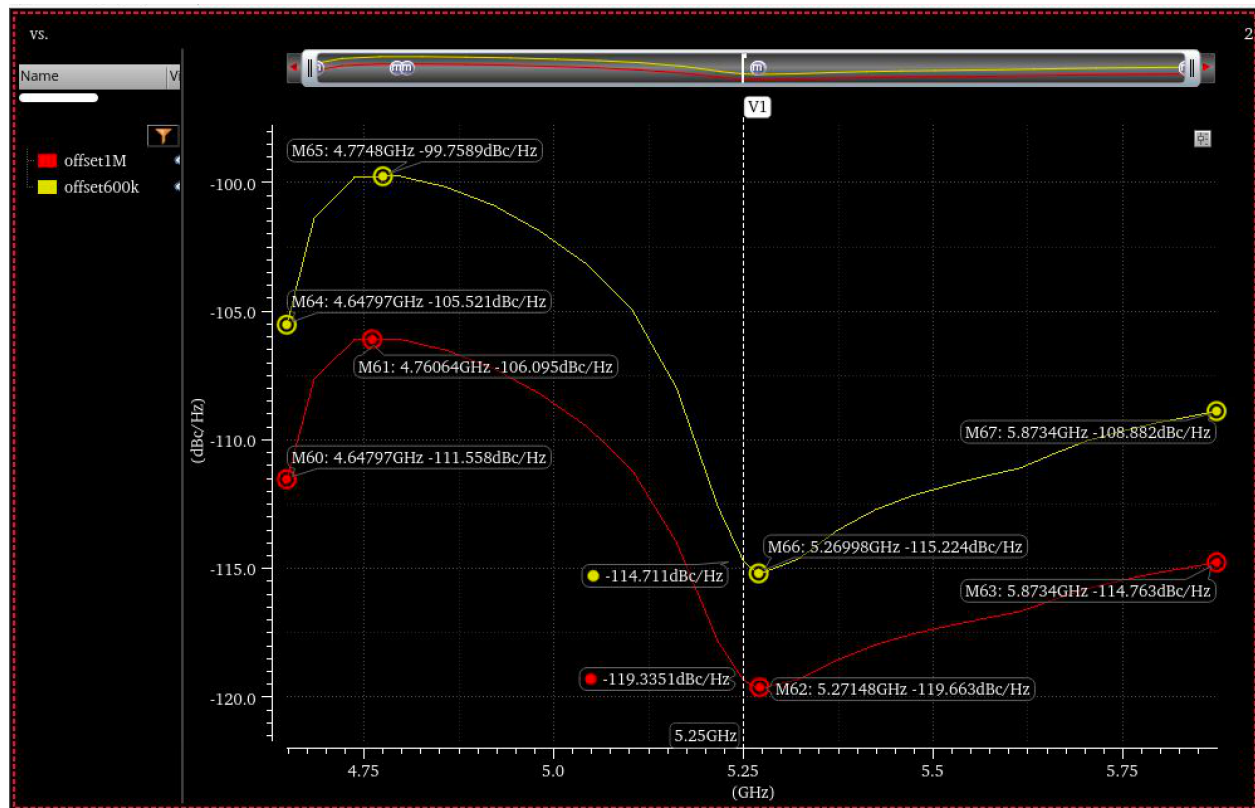


Fig 10. Spot Noise at 600 kHz (yellow) and 1 MHz (red) Offsets vs Tuned Frequency

A Phase Noise summary was produced to identify major sources of noise inside the VCO.

Device	Param	Noise Contribution	% Of Total
/I6/NM0	id	5.88702e-13	17.50
/I6/NM1	id	5.88575e-13	17.49
/I6/NM1	fn	5.12509e-13	15.23
/I6/NM0	fn	5.12442e-13	15.23
/I6/NM4	id	2.69792e-13	8.02
/I6/I0/R0	rn	1.6671e-13	4.96
/I6/I1/R0	rn	1.66684e-13	4.95
/I6/I4/NM0/xrg/r1	thermal_noise	1.6417e-13	4.88
/I6/I5/NM0/xrg/r1	thermal_noise	1.64144e-13	4.88
/I6/NM1/xrg/r1	thermal_noise	5.00899e-14	1.49

Spot Noise Summary (in V²/Hz) at 1M Hz Sorted By Noise Contributors
Total Summarized Noise = 3.36438e-12
No input referred noise available
The above noise summary info is for pnoise_pm data

Device	Param	Noise Contribution	% Of Total
/I6/NM1	fn	2.05034e-12	19.33
/I6/NM0	fn	2.05002e-12	19.33
/I6/NM0	id	1.63528e-12	15.42
/I6/NM1	id	1.63493e-12	15.42
/I6/NM4	id	7.49422e-13	7.07
/I6/I0/R0	rn	4.63082e-13	4.37
/I6/I1/R0	rn	4.63012e-13	4.37
/I6/I4/NM0/xrg/r1	thermal_noise	4.56028e-13	4.30
/I6/I5/NM0/xrg/r1	thermal_noise	4.55957e-13	4.30
/I6/NM1/xrg/r1	thermal_noise	1.39139e-13	1.31

Spot Noise Summary (in V²/Hz) at 600K Hz Sorted By Noise Contributors
Total Summarized Noise = 1.06048e-11
No input referred noise available
The above noise summary info is for pnoise_pm data

Fig 11. Phase Noise Summary

III. DESIGN METHODOLOGY AND FIRST PASS CALCULATIONS

A. Minimizing Phase Noise

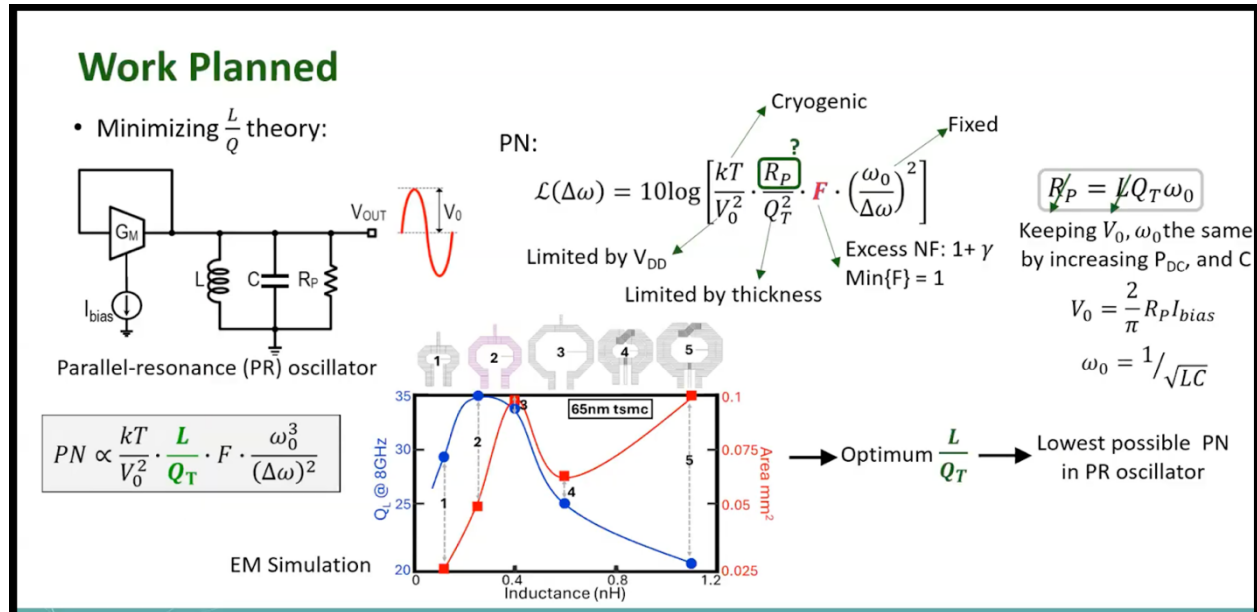


Fig. 12 ELEC 404 Tutorial Slide

In order to minimize phase noise, we want to minimize L/Q . The Pi inductor of the LC tanks of our VCO control Q and L so we want to maximize Q/L which we can find from a SP analysis sweep from 0.2n to 2n for L.

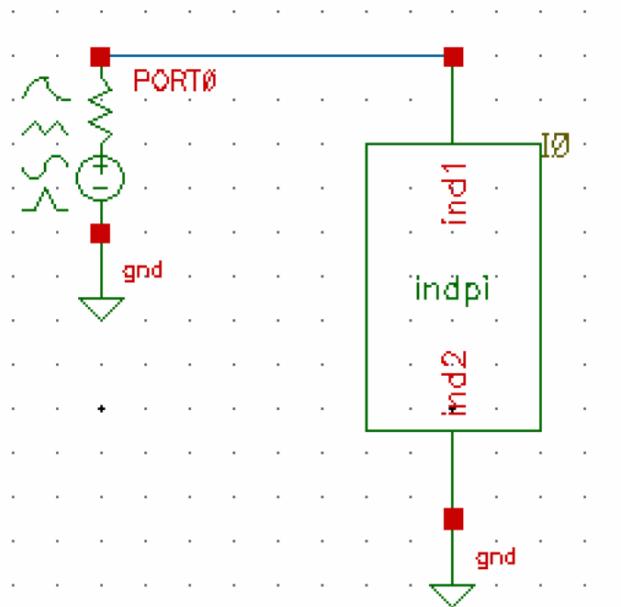


Fig 13. Pi Model Inductor Testbench

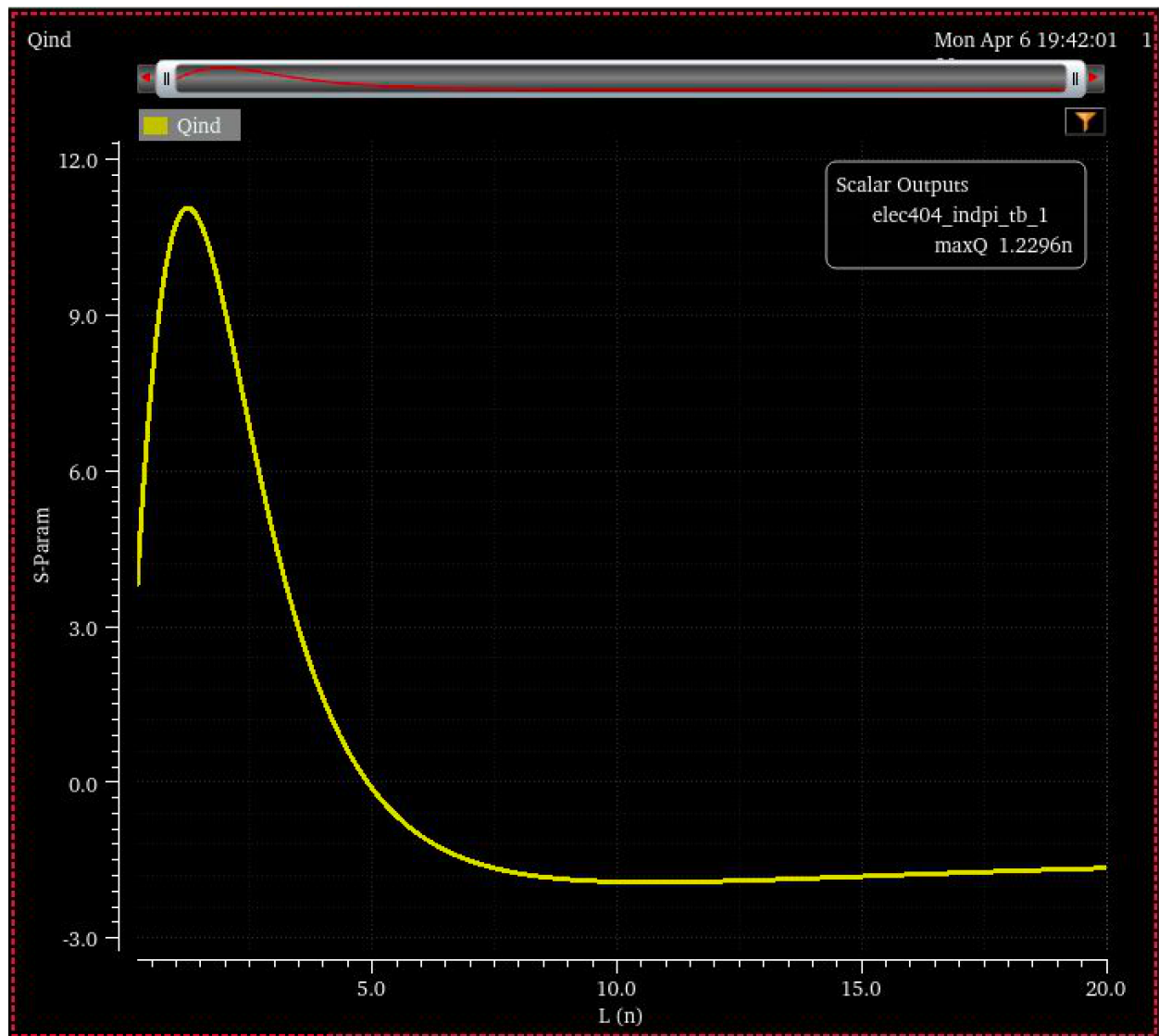


Fig 14. Qind vs L

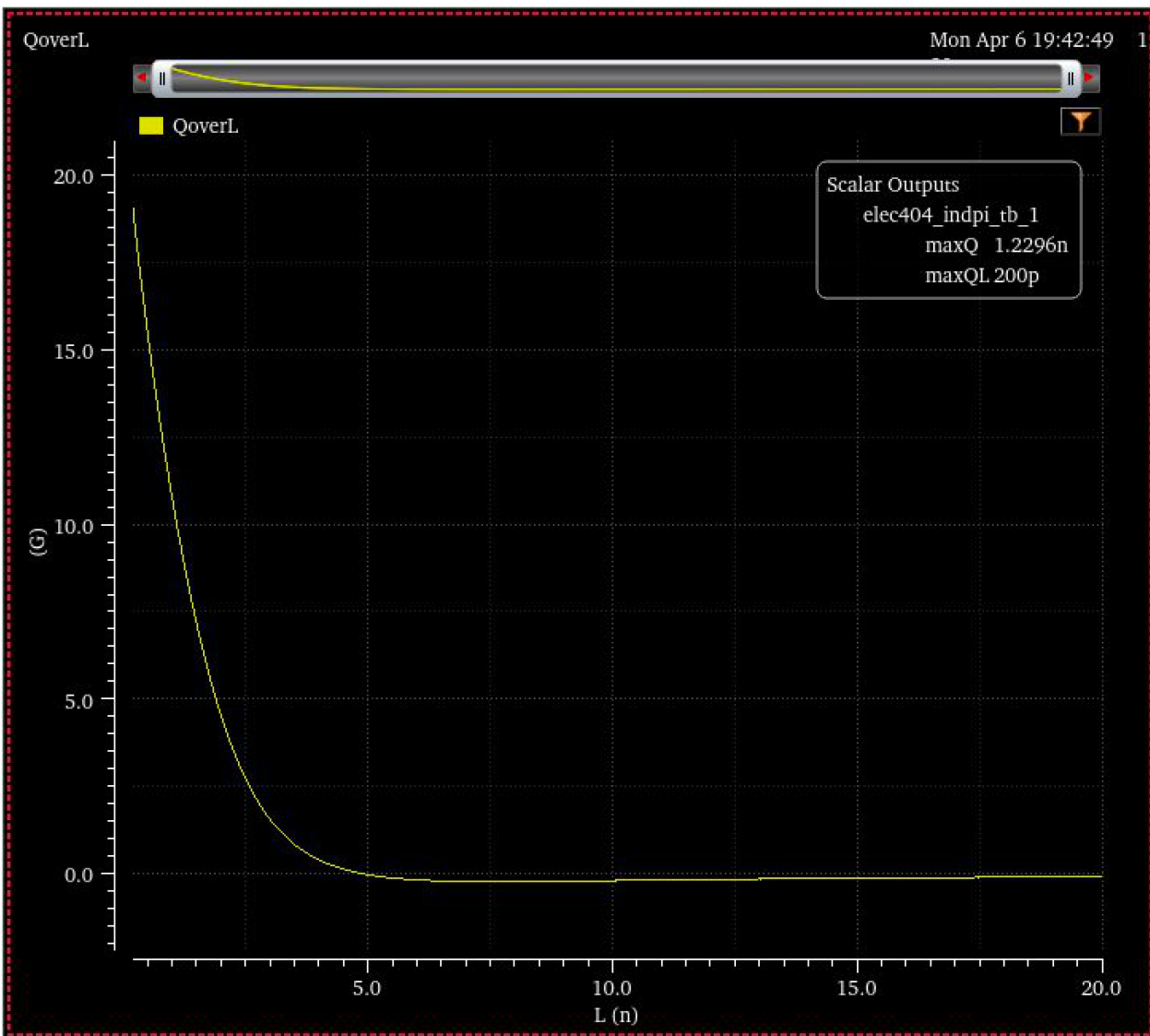


Fig 15. Q/L vs L

Q/L is maximized at 200p which seemed super low. Q was maximized at $L = 1.23$ nH which seems more realistic. For that value of L we have $R_p = 513$ Ohm we can estimate with 500 Ohm to start with. The center frequency is 5.25 GHz $= 1/\sqrt{LC}$ which let's us solve for the total $C = 1/L/(2\pi \cdot 5.25\text{GHz})^2 = 766$ fF which we can estimate to be about 750 fF. This total capacitance is C_{load} (80 fF) + C_{add} (some constant to meet frequency requirements) + C_{par} (of the cross coupled transistors) + C_{var} (of the varactor). The tuning range specification is 10% so $f_{\text{min}} = 4.725$ GHz and $f_{\text{max}} = 5.775$ GHz which correspond to total capacitance ranging from 945 fF to 633 fF. This means the varactor capacitance must be about 320 fF at least.

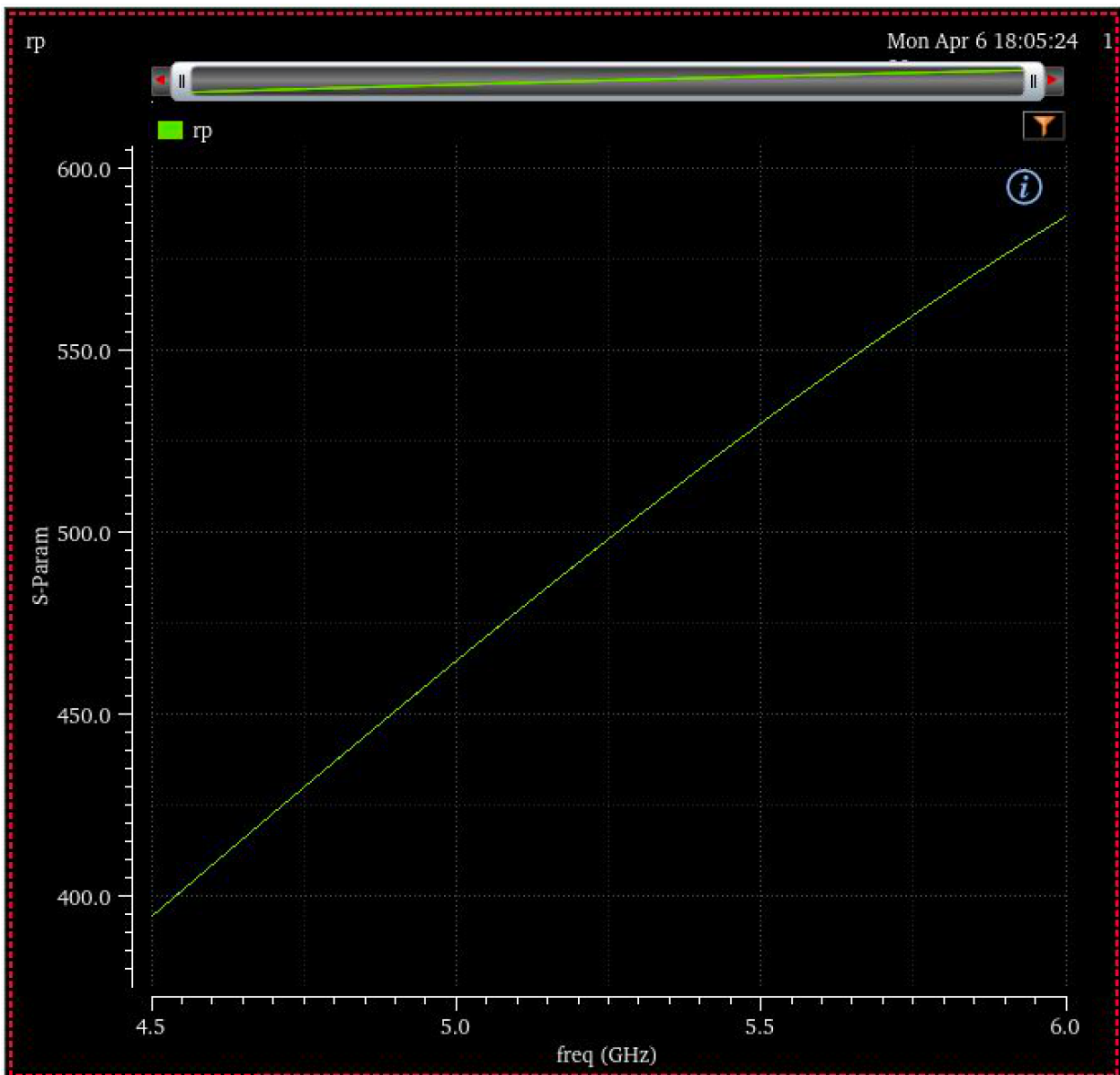


Fig. 16 Rp (for L = 1.2 nH) vs Operating Frequency

As shown in Figure 16, we can sweeping frequency across the tuning range at L = 1.2 nH and find that Rp ranges from about 400 Ohms to 600 Ohms. Rp is about 500 Ohms at the center frequency of 5.25 GHz.

Now we also want an idea of the parasitic capacitances of the cross coupled transistor. Since $C_{par} = C_{gs} + C_{db} + 4 \cdot C_{gd}$, we can create a testbench to figure out the unit parasitic capacitance is approximately 28.29 fF/(μm)².

Name	Type	Details	Value
sat	expr	OP("/NM0" "region")	2
cgs	expr	OP("/NM0" "cgs")	-6.723f
cdb	expr	OP("/NM0" "cdb")	-302.4z
cgd	expr	OP("/NM0" "cgd")	-1.502f
cpar	expr	abs((cgs + cdb + (4 * cgd)))	12.73f
df/dum^2	expr	(cpar / (VAR("l") / 1e-06) / (VAR("w") / 1e-06))	28.29f

Fig. 17 Unit Parasitic Capacitance DC Simulation

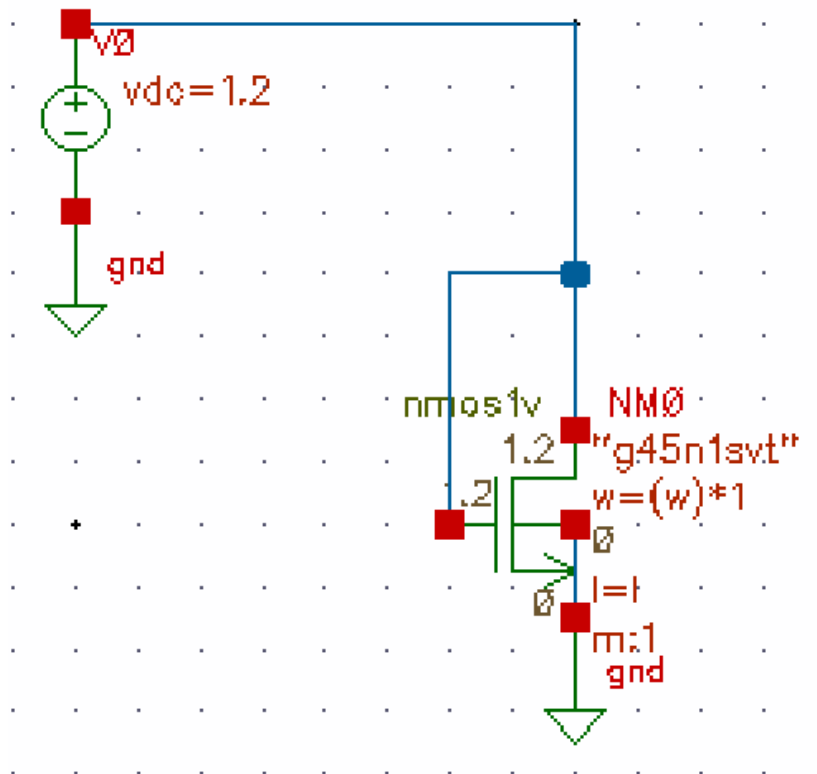


Fig. 18 Parasitic Capacitance Testbench

We also need to identify transistor sizings that create a suitable capacitance range for the varactor.

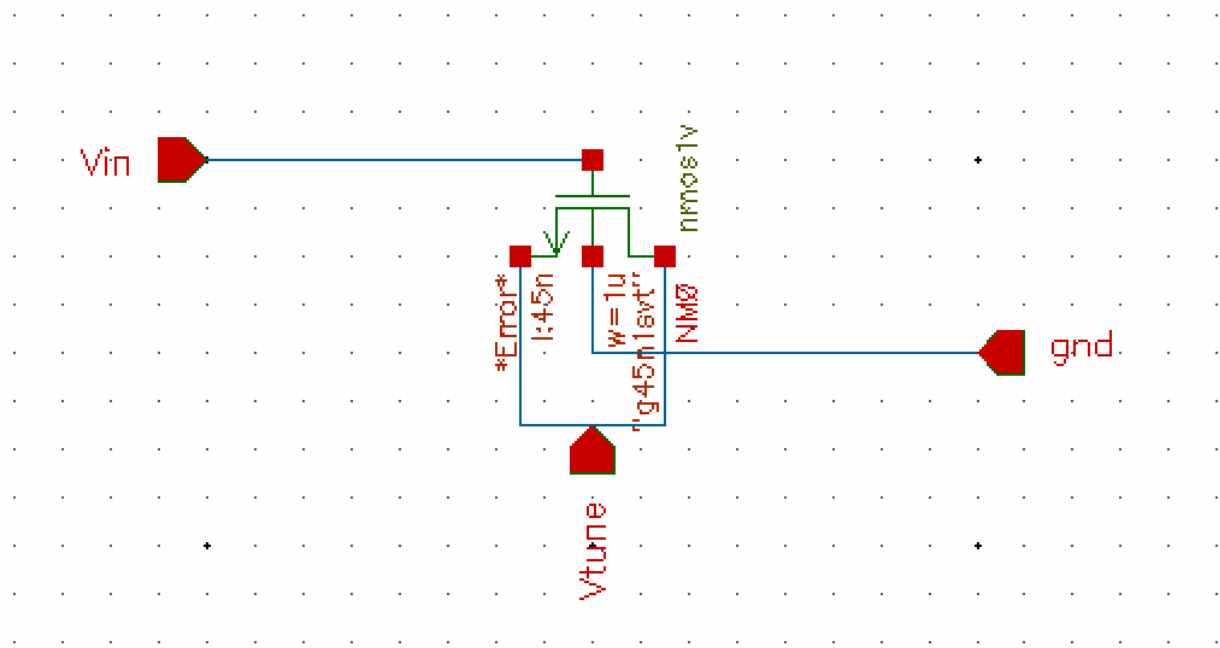


Fig. 19 Varactor Schematic

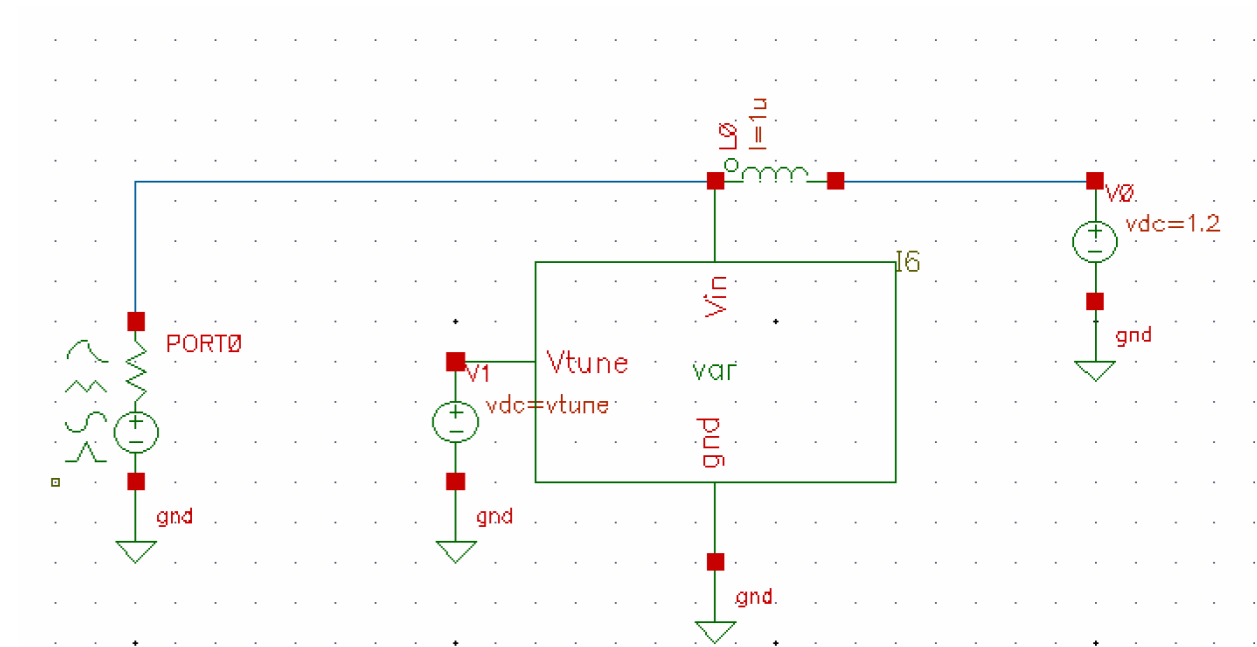


Fig. 20 Varactor Testbench

Using a width of 10 μm , length of 1 μm , and multiplier of 5 we can find adequate first pass values for the varactor. The capacitance range is 70 fF to 665 fF which allows for a range of approximately 600 fF

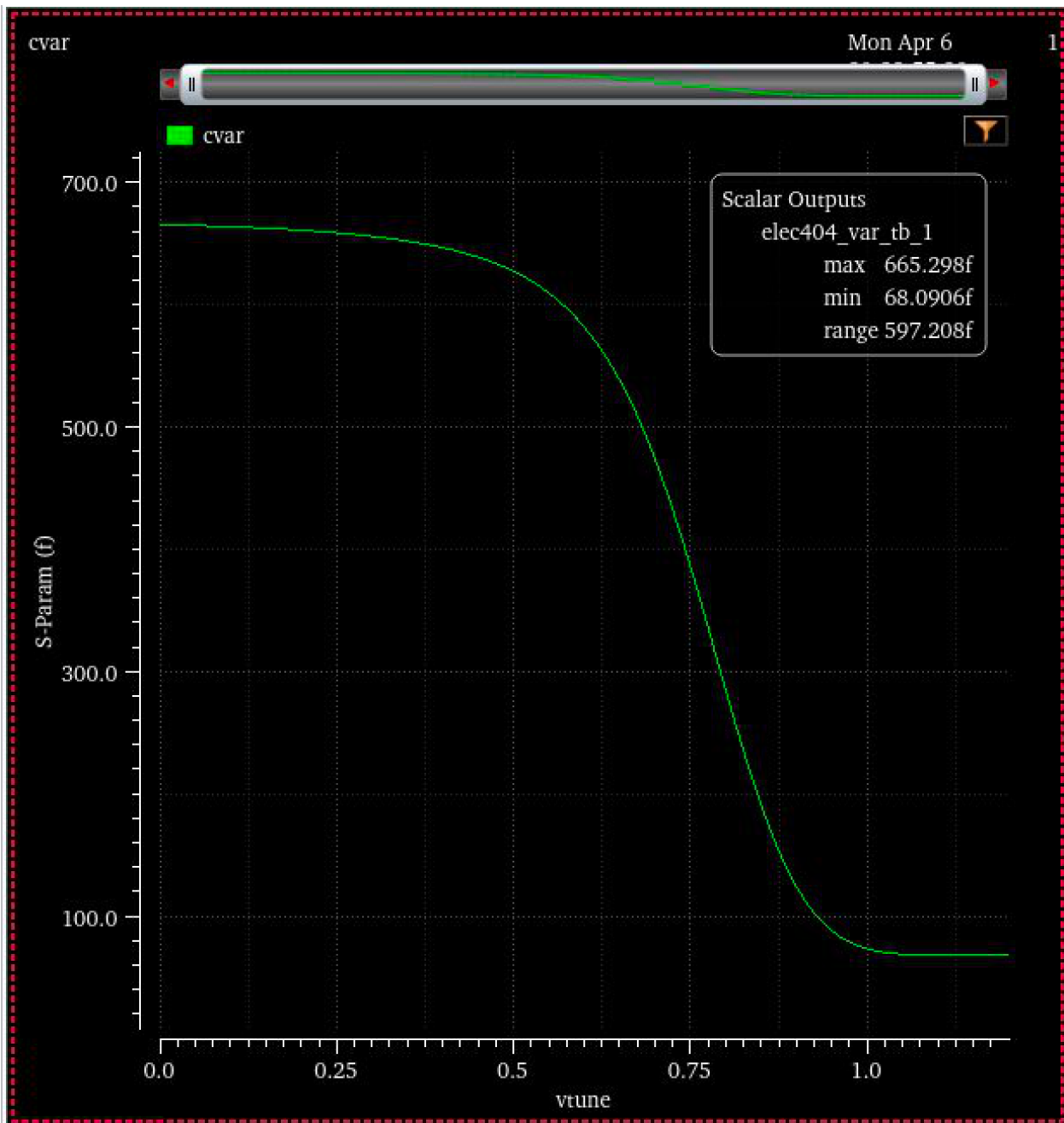


Fig. 21 Varactor Capacitance vs tuning voltage (V_{tune})

In order for the VCO to function correctly we have a startup condition that $g_m \cdot R_p \gg 1$. With our R_p starting at around 400 Ohms, this means $g_m > 1/400 = 2.5$ mS. However to be safer we can require g_m to be larger so we want $g_m > 5$ mS. Sweeping across various lengths, widths, and multipliers for the transistor with $L = 1.2$ nH we find that $I_{\text{bias}} = 5$ m, $\text{len} = 45$ n, $\text{width} = 1$ u, $\text{mult} = 10$ works well enough as $g_m = 10.37$ mS.

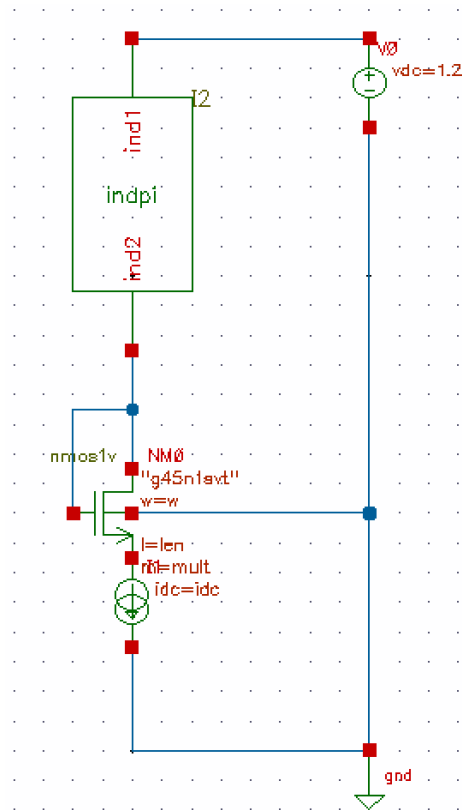


Fig. 22 Cross Coupled Transistor Test Bench

Then we have a set of first pass values based on our hand calculations that we can begin fine tuning to meet the VCO spec through simulation.

B. First Pass Values

TABLE 3
First Pass Values found from Hand Calculations

L_indpi_0,1	1.2nH
Ibias	2*5mA
L_0,1	45nm
L_4,5	2*45nm
L_var	1um
M_0,1	10
M_4,5	10
M_var	5
W_0,1	1um
W_4,5	2*1um
C_par	$30\text{fF}/1\text{um}^2 \cdot 0.45 = 13.5\text{ fF}$
C_load	80 fF
C_var	70 to 670 fF
C_add	$600 - 13.5 - 80 - 70 = 450\text{ fF}$

IV. POWER CONSUMPTION OPTIMIZATION

A. Initial Power Consumption

The initial design consumed only 12 mW of power, but did not meet any specifications. Making the bias current super large helped meet the specification as it reduced noise and helped with the tuning range, however this was not very good for power consumption. Thus, other techniques were needed to increase the tuning range and reduce phase noise while consuming much less power.

B. Optimization Techniques

Several techniques were employed to minimize power consumption:

- Use minimum required negative resistance
 - Power consumption was reduced by sizing the cross-coupled NMOS pair to provide only the minimum required negative resistance for startup. A startup margin of approximately 2 was selected to ensure reliable oscillation while avoiding excessive bias current.
- Increase tank Q before increasing current
 - Rather than increasing tail current to improve phase noise, the LC tank quality factor was improved through inductor metal optimization and by allocating more of the total capacitance budget to low-loss MIM capacitors
- Tail Current Source Sizing
 - To reduce power while preserving voltage swing under a 1.2 V supply, the lengths of the tail current mirror were increased. This removes headroom limitations and suppresses excess flicker noise upconversion from the tail current source.
- Use transistor sizing as power optimization
 - Device widths were iteratively optimized to maximize gm/Id while limiting parasitic capacitance loading on the resonant tank.

C. Final Power Consumption

Through the iterative optimization, the final design achieved 15.3 mW power consumption

V. CONCLUSION

A CMOS voltage controlled oscillator operating at a center frequency of 5.25 GHz was successfully designed, analyzed, and simulated using 45 nm PDK models with a 1.2 V supply voltage. The LC cross-coupled topology together with a voltage-controlled MOS varactor network enabled the oscillator to achieve all required specifications while maintaining low power operation. The final design achieved a tuning range from 4.65 GHz to 5.79 GHz, which exceeds the required 10% tuning range specification. The measured overall VCO gain was 948 MHz/V, with an even local linear-region gain of approximately 1.24 GHz/V.

The phase-noise performance also exceeded the design targets, achieving -114.6 dBc/Hz at 600 kHz offset and -119.3 dBc/Hz at 1 MHz offset, satisfying the low-noise local oscillator. These results were achieved through systematic hand analysis, first-pass values for LC tank and oscillation startup calculations, and iterative RF simulation using PSS and Pnoise analyses in Cadence Spectre RF. Particular attention was devoted to minimizing power consumption through startup margin optimization, resonator quality factor improvement, careful tail current source sizing, and transistor parasitic capacitance management.

The final design consumed only 15.3 mW, demonstrating that GHz-range VCO phase-noise and tuning specifications can be achieved in advanced CMOS technology without excessive power dissipation. Overall, this work demonstrates that high-performance low-power CMOS VCOs suitable for 5.25 GHz wireless local oscillator applications can be successfully realized using careful LC tank optimization and transistor sizing techniques.